First Name	Last Name

Before exam begins:

- Write your first name and last name in the boxes above.
- Turn off all electronics and keep them out of sight: no cellular phones, not even to tell time (and not even if it's just in airplane mode). No iPods, wearing of headphones, etc.
- The cover page should only have your name. **NOTHING ELSE MAY BE WRITTEN ON THE COVER PAGE.**

As soon as exam starts:

- Take a quick breath to relax!!
- Check that you have all 9 pages of the exam. (The number of pages includes this cover page.)
- Scratch paper will be available. Use the provided scratch paper.

During the exam:

- **TO EARN POINTS YOU MUST SHOW YOUR WORK**. This means you should explain your reasons and justify your steps. Your work should be written in a clear and logical manner. It is important to show me that you understand the process, showing a reasonable number of steps.
- Keep your eyes on your own exam!
- **No calculators**, phones, or other electronics at all. They should be turned off!
- No notes, books, or other outside resources.
- If you have left over time, check your work!

The exam length is exactly 55 minutes. There are 100 maximum possible points.

- The exam starts when you have received confirmation from the instructor, **NOT** as soon as you receive the exam. Likewise, when you are told to stop, you must stop **IMMEDIATELY**. This is in fairness to all students. Do not think that you are the exception to this rule. Continuing to work on the exam after time has been called will have serious consequences.

1. Define each of the following terms:

(a) Define subset.

Say S and T are sets. Then S is a subset of T if every element of S is an element of T.

This is written $S \subseteq T$.

(b) Define divides.

Say s and t are integers. Then s divides t if there exists $m \in \mathbb{Z}$ such that $sm = t$.

(c) Define Cartesian product.

Given sets A and B, the Cartesian product $A \times B$ of A and B is the set $\{(c,d) : c \in A, d \in B\}$

alternate: $\{z \mid \exists c \in A \ \exists d \in B \text{ s.t. } z = (c,d)\}$

2. Define each of the following terms:

(a) Define odd.

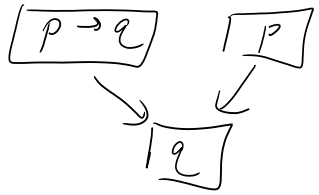
Say k is an integer. We say k is odd if $\exists z \in \mathbb{Z}$ s.t. $k = 2z + 1$.

(b) State modus ponens.

Either

When we know an Implication is true and we also know the "if" part of this Implication, we get to conclude the "then" part.

Diagram okay too!



(c) State one of De Morgan's Laws.

Either

$\neg(p \wedge q) \equiv (\neg p) \vee (\neg q)$

$\neg(p \vee q) \equiv (\neg p) \wedge (\neg q)$

3. If you know $\boxed{\text{For all } y \in G, \text{ there exists } m \in M \text{ such that } m \text{ gives } y \text{ a gift}}$ and you know $\boxed{y \in M}$ what (if anything) can you conclude? Briefly explain.

We cannot conclude anything.

Explanation: To use $\boxed{\forall y \in G \ldots}$ we need to already know of something in the set G.

4. If you know $\boxed{\text{If } x \text{ likes to swim, then } y \text{ has rollerblades}}$ and you know $\boxed{y \text{ does not rollerblade}}$ what (if anything) can you conclude? Briefly explain.

We can conclude $\boxed{x \text{ does not like to swim}}$

Explanation: Modus tollens

5. What do you have to do to prove $\boxed{\text{For all } x \in C, \ x \text{ wears shoes}}$?

Start by writing $\boxed{\text{Let } x \in C \text{ be arbitrary}}$

Note: you cannot say $\boxed{\text{Prove } x \in C}$ (:(

Then, work to prove $\boxed{x \text{ wears shoes}}$

6. If the set Z is defined to be $Z = \{T(v) : v \in V\}$ what do you have to do to prove $\boxed{a \in Z}$?

$Z = \{n \mid \exists v \in V \text{ s.t. } n = T(v)\}$

Prove $\exists v \in V$ s.t. $a = T(v)$.

$\exists v \in V$ s.t. $a = T(v)$

↓

$a \in Z$

7. For both parts below, the set M is defined $M = \{\sin(x) : x \in L\} = \{v \mid \exists x \in L \text{ s.t. } v = \sin(x)\}$

(a) When given the info $\boxed{h \in M}$ what can we conclude?

↓

Conclude $\boxed{\exists x \in L \text{ s.t. } h = \sin(x)}$

⚠ It is incorrect to conclude $h \in L$

(b) When given the info $\boxed{c \in L}$ what can we conclude?

Conclude $\sin(c) \in M$.

Why? Say we let $r = \sin(c)$. Then, my claim is $r \in M$.

Based on (a) replacing h there work's

$\exists c \in L$ s.t. $r = \sin(c)$ is true.

8. Use the following set definitions for all parts below. Let $A = \{6, 8, 10, 12\}$ and $B = \{5, 7\}$ and $C = \{x \in \mathbb{Z}_{>0} \mid x \text{ is even and } x < 7\}$ and $D = \{3x : x \in \mathbb{Z}\}$. $= \{\ldots, -9, -6, -3, 0, 3, 6, \ldots\}$

(a) Write $B \times C$ in roster form. $C = \{2, 4, 6\}$

$B \times C = \{(5,2), (5,4), (5,6), (7,2), (7,4), (7,6)\}$

(b) State two elements of $A \times D$ with proper notation.

$(8, 3) \in A \times D$ ← notation! ← one element of $A \times D$

$(12, -9) \in A \times D$ ← another element of $A \times D$

Note: each element of $A \times D$ is an ordered pair!

(c) Write the power set of C in roster form.

$P(C) = P(\{2,4,6\}) = \{\emptyset, \{2\}, \{4\}, \{6\}, \{2,4\}, \{2,6\}, \{4,6\}, \{2,4,6\}\}$

$\emptyset$ can be replaced with $\{\}$ but not with $\{\emptyset\}$

(d) What is the cardinality of $A \times C$?

$$|A \times C| = |A| \times |C|$$
$$= 4 \cdot 3$$
$$= 12$$

9. Given the set definitions:

- $X = \{c \in G : c \text{ is a pastry chef and } c \in S\}$
- $Y = \{c \in H \mid c \text{ snowboards and } c \text{ is a reindeer}\}$
- $S = \{h \in H : h \text{ likes warm hugs}\}$

Use the hypotheses:

- $\forall x \in X$, if x likes warm hugs, then x is a reindeer.
- $G \subseteq K$
- $\forall k \in K$, if k does not snowboard, then k is not a pastry chef.

prove that $X \subseteq Y$.

Suppose $m \in X$. WTS $m \in Y$. Since $m \in X$, by H1, the implication "if m likes warm hugs, then m is a reindeer" is true. Since $m \in X$, we know $m \in G$ and we know m is a pastry chef and $m \in S$. Since $m \in S$, we get $m \in H$ and m likes warm hugs. Applying our earlier implication to m likes warm hugs, we conclude m is a reindeer.

Since $m \in G$, by the second hypothesis, $m \in K$. Since $m \in K$, from H3, we learn that m does not snowboard implies m is not a pastry chef. But recall m is a pastry chef, so by modus tollens m snowboards.

We have proved $m \in H$ and m snowboards and m is a reindeer, which proves $m \in Y$.

Since we proved $m \in X$ implies $m \in Y$, we have proved $X \subseteq Y$. □

10. Given the definitions:

- An integer c is *threeven* if there is an integer b such that $3b = c$.
- An integer c is *trodd* if there is an integer b such that $3b + 1 = c$.
- An integer c is *trodder* if there is an integer b such that $3b + 2 = c$.

prove the statement:

- For all $a \in \mathbb{Z}$ and for all $b \in \mathbb{Z}$, if a is trodder and b is trodder, then $a + b$ is trodd.
  [Note: a and b are <u>both</u> trodder]

Let $a, b \in \mathbb{Z}$ be arbitrary. Suppose a is trodder and b is trodder.

We want to show $a+b$ is trodd. Since a and b are trodder, there exist integers m and n s.t. $a = 3m+2$ and $b = 3n+2$. Let $s = m+n+1$. Then $s \in \mathbb{Z}$ and $a+b = 3m+2+3n+2 = 3m+3n+4 = 3m+3n+3+1 = 3(m+n+1)+1 = 3s+1$.

So $a+b$ is trodd.

11. Prove: For all $a \in \mathbb{R}$, for all $b \in \mathbb{R}$, for all $c \in \mathbb{R}$, if $5 \mid (a-b)$ and $5 \mid (b-c)$, then $5 \mid (a-c)$.

Let $a \in \mathbb{R}$. Let $b \in \mathbb{R}$. Let $c \in \mathbb{R}$. WTS if $5 \mid (a-b)$ and $5 \mid (b-c)$, then $5 \mid (a-c)$.

Assume $5 \mid (a-b)$ and $5 \mid (b-c)$. WTS 5 divides $a-c$. Since 5 divides $a-b$, there exists $x \in \mathbb{Z}$ s.t. $5x = a-b$. For similar reasons, there exists $y \in \mathbb{Z}$ s.t. $5y = b-c$.

Let $u = x+y$. Then $u \in \mathbb{Z}$ and $5u = 5(x+y) = 5x+5y = (a-b)+(b-c) = a-b+b-c = a-c$, which proves 5 divides $a-c$. □

THIS IS THE LAST PAGE OF THE EXAM.